Algebra Inequalities Cheat Sheet

Unit Focus: Solving, Graphing, and Writing Inequalities

1. What Is an Inequality?

An **inequality** compares two quantities that may not be equal.

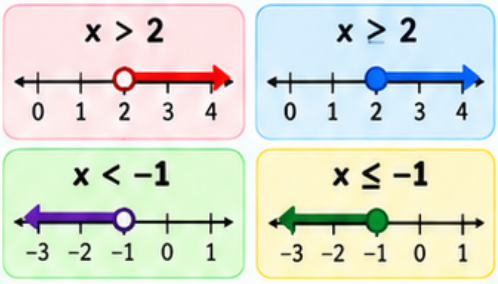
Symbol	Meaning	Example
$<$	less than	$3 < 5$
$>$	greater than	$x > 2$
$\leq$	less than or equal to	$y \leq 7$
$\geq$	greater than or equal to	$n \geq -1$
$\neq$	not equal to	$a \neq 4$

2. Vocabulary & Key Phrases

Phrase	Symbol
less than	$<$
greater than	$>$
at most	$\leq$
at least	$\geq$
no more than	$\leq$
no less than	$\geq$
fewer than	$<$
more than	$>$
is not equal to	$\neq$

3. How to Graph on a Number Line

- **Open circle:** use for $<$ or $>$ (values not included).
- **Closed circle:** use for $\leq$ or $\geq$ (values included).
- Shade **left** for values less than.
- Shade **right** for values greater than.



4. How to Solve an Inequality

1. Solve it like an equation using inverse operations.
2. Keep the variable on one side.
3. Check your answer when needed.

IMPORTANT: If you multiply or divide by a negative number, **reverse the inequality symbol.**

5. Examples

A. One-step: Addition

Solve: $x + 4 > 9$

$x + 4 > 9$

$-4 \quad -4$

$x > 5$

x > 5

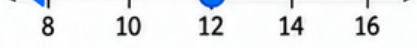


B. One-step: Division

Solve: $\frac{m}{3} \leq 4$

m ≤ 12

m ≤ 12



C. Reverse-Sign Example

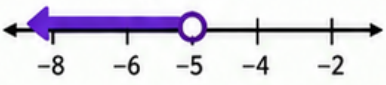
Solve: $-2p > 10$

$-2p > 10$

$\div(-2) \quad \div(-2)$

$p < -5$

p < −5



D. Two-step: Addition and Division

Solve: $3x - 7 \leq 8$

$3x - 7 \leq 8$

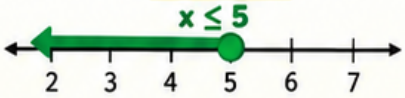
$+7 \quad +7$

$3x \leq 15$

$\div 3 \quad \div 3$

$x \leq 5$

x ≤ 5



6. Check a Solution

For $x + 4 > 9$, test $x = 6$:

$6 + 4 > 9$

$10 > 9$ ✓ **true**

So $x = 6$ is a solution.

Check $x = 4$ (this would not work):

$4 + 4 > 9$

$8 > 9$ ✗ **false**

So $x = 4$ **is not a solution.**

7. Common Mistakes

- Forgetting to reverse the symbol after multiplying or dividing by a negative.
- Using the wrong circle on the graph.
- Shading the wrong direction.
- Not isolating the variable completely.

Check your work!

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More Practice: Writing, Solving, and Compound Inequalities

1. Important Vocabulary

Term	Definition
solution	a value that makes an inequality true
solution set	all values that make the inequality true
boundary point	the endpoint shown by the circle on the graph
compound inequality	two inequalities joined by "and" or "or"
inequality symbol	$<$, $>$, $\leq$, or $\geq$

2. Write an Inequality from Words

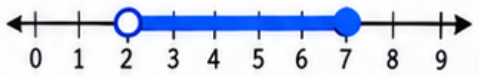
Phrase	Inequality
a number x is at least 6	$x \geq 6$
a number n is less than 12	$n < 12$
7 more than y is no more than 19	$y + 7 \leq 19$
twice a number p is greater than 14	$2p > 14$

Watch out: "less than" can reverse the order in a phrase. "5 less than x" means $x - 5$, not $5 - x$.

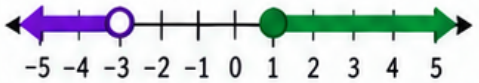
3. Compound Inequalities

- **AND** means **both** inequalities are true. Graph the overlap (the values that satisfy both).
- **OR** means **either** inequality is true. Graph both parts.

A. AND Example: $2 < x \leq 7$



B. OR Example: $x < -3$ or $x \geq 1$



4. Solve a Compound Inequality

A. $3 < x + 2 \leq 8$

Subtract 2 from all three parts.

$3 - 2 < x + 2 - 2 \leq 8 - 2$

$1 < x \leq 6$

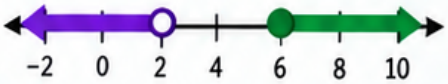


B. $2x - 1 < 5$ or $x + 4 \geq 10$

Solve each side.

$2x - 1 < 5$	$x + 4 \geq 10$
$2x < 6$	$x \geq 6$
$x < 3$	

Final answer: $x < 3$ or $x \geq 6$



5. Variables on Both Sides

Solve: $5x + 3 > 2x + 12$

Subtract $2x$ from both sides.

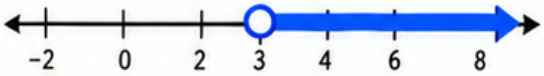
$3x + 3 > 12$

Subtract 3 from both sides.

$3x > 9$

Divide by 3 (positive, so no reversal).

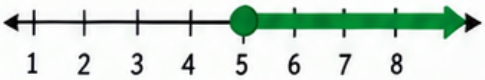
$x > 3$



6. Use the Distributive Property

Solve: $2(x - 1) \geq 8$

Distribute.	$2x - 2 \geq 8$
Add 2 to both sides.	$2x \geq 10$
Divide by 2 (positive).	$x \geq 5$



7. Special Cases

A. No Solution

Solve: $2x + 1 < 2x - 4$

Subtract $2x$ from both sides.

$1 < -4$

false

So there is **no solution.**

B. All Real Numbers

Solve: $4(x + 1) \geq 4x + 1$

$4x + 4 \geq 4x + 1$

Subtract $4x$ from both sides.

$4 \geq 1$

true

So all real numbers are solutions.

8. Quick Reminders

1. Use **open** circles for $<$ or $>$.
2. Use **closed** circles for $\leq$ or $\geq$.
3. Shade **left** for less than.
4. Shade **right** for greater than.
5. **Reverse** the inequality when multiplying or dividing by a negative.
6. Check whether your answer makes sense.

Check your work!

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